

vivainsights R library: CHEAT SHEET (1/2)

Full documentation site:
<https://microsoft.github.io/vivainsights/>



Basics

 **vivainsights** is an R library that offers a set of tools and functions for analysing and visualising data from [Microsoft Viva Insights](#)

You can install or update the package with the following command from CRAN: `install.packages("vivainsights")`

 The latest development version and documentation can be found on our GitHub repository: <https://github.com/microsoft/vivainsights/>

?function_name

To load the documentation in R, prefix the function name with a question mark

Tip!
Load **{tidyverse}** as the companion package to **Viva Insights** for seamless data wrangling

Example set up

```
library(vivainsights)
library(tidyverse)
pq_df <- import_query("path/...csv")

# Collaboration hours - summary
collab_sum(pq_df)
```

Data import / export

Use our handy functions below which are optimized for best practice for getting data *in* and *out* of R

- import_query()**
Import CSV queries faster and pre-formatted for **vivainsights** functions (instead of `read.csv()`)
- export()**
Copy a data frame to clipboard, or write as a CSV, or a ggplot object as PNG or SVG

Data validation

Validate and understand your data prior to starting a piece of analysis

	validation_report() Generate a report to validate person query data, with options to supply an additional <i>meeting query</i>
	hrvar_count() Count number of employees in HR attribute
	extract_hr() Extract HR attributes in a query
	check_query() Print diagnostic data about the query to the R console
	identify_holidayweeks() Identify likely holiday weeks (for the entire pop) where collaboration hours lie far outside the mean
	identify_nkw() Identify likely non-knowledge workers where average person collaboration hours lie far outside the mean
	identify_inactiveweeks() Identify likely person-weeks where collaboration hours lie far outside the mean relative to the population average
	identify_tenure() Calculate tenure based on a supplied hire date
	identify_outlier() Takes in a selected metric and uses z-score (number of standard deviations) to identify outliers across time
	identify_privacythreshold() This function scans a standard query output for groups with of employees under the privacy threshold
	hr_trend() Returns a line chart showing the change in employee count over time
	hrvar_trend() Tracks distinct employee counts over time for a chosen HR attribute (e.g., org, level) and returns a trend plot.
	identify_churn() Flags employees who have “churned” (dropped out) between time periods based on presence/absence across dates.
	identify_usage_segments() Classifies people into usage segments based on a selected metric (passed as a metric/column)

Inbuilt datasets

Explore **vivainsights** by using inbuilt demo datasets

	pq_data Person query
	mt_data Meeting query
	p2p_data Person-to-Person query
	p2p_data_sim() Person to Person query / edge list based on the graph (Watts-Strogatz small-world network model)
	g2g_data Loads a sample group-to-group (edge-list) dataset for network analysis demos and examples

Exploratory analysis

Explore the data and surface initial hypotheses

	keymetrics_scan(), keymetrics_scan_asis() Returns a heatmapped table by default, with options to return a table; Return a heatmapped table directly from the aggregated / summarized data
	create_rank(), create_rank_combine() Returns a plot by default, with an option to return a table with all groups (across multiple HR attributes) ranked by the specified metric

Distribution

Understand the distribution of a metric

	create_boxplot() Analyzes a selected metric and returns a box plot by default
	create_density() Returns a faceted density plot by default
	create_hist() Returns a faceted histogram by default

Basic analysis

Combine prefix with plot type to create a specific analysis on a Viva Insights metric
Available prefixes: **collab, email, meeting, afterhours, one2one, workloads**

	*_summary() <i>hrvar, mingroup, return</i> Returns a bar plot showing average weekly email hours by default
	*_dist() <i>hrvar, mingroup, return</i> Returns a stacked bar plot by default
	*_fizz() <i>hrvar, mingroup, return</i> Returns a 'fizzy' scatter plot by default
	*_line() <i>hrvar, mingroup, return</i> Returns a line chart for email hours by default
	*_trend() <i>hrvar, mingroup, return</i> By default, returns a week-by-week heatmap, highlighting the time with most activity
	*_rank() <i>hrvar, mingroup</i> Returns a plot by default, with an option to return a table with a all of groups (across multiple HR attributes) ranked by hours of digital collaboration





Flexible analysis

Flexible analysis functions are versatile, allowing you to pass any metric as a string parameter, e.g., **metric = 'Email_hours'**

 **create_bar(), create_bar_asis()**
metric, hrvar, mingroup, return
Returns a bar plot showing the average of a selected metric by default. This function creates a bar chart directly from the aggregated / summarized data

 **create_fizz()**
metric, hrvar, mingroup, return
Analyzes a selected metric and returns a 'fizzy' scatter plot by default

 **create_scatter()**
metric, hrvar, mingroup, return
Returns a scatter plot of two selected metrics, using color to map an HR attribute

 **create_bubble()**
metric, hrvar, mingroup, return
Returns a bubble plot of two selected metrics, using size to map the number of employees

 **create_dist()**
metric, hrvar, mingroup, return
Returns a stacked bar plot by default

 **create_inc()**
metric, hrvar, mingroup, return
Returns a heatmap for the generated incidence analysis

 **create_sankey()**
data, var1, var2, count
Create a 'networkD3' style sankey chart based on a long count table with two variables

 **create_stacked()**
metric, hrvar, mingroup, return
Returns a stacked bar plot by default

Flexible analysis - over time

Flexible analysis functions for understanding changes over time

 **create_line(), create_line_asis()**
hrvar, mingroup
By default, returns a line chart for the defined metric

 **create_period_scatter()**
metric_x, metric_y, hrvar, mingroup, return
Returns a faceted scatter plot by default

 **create_trend()**
metric, hrvar, mingroup, return
By default, returns a week-by-week heatmap bar plot, highlighting the points in time with most activity

 **create_tracking()**
metric, percent
Create a line chart that visualizes a set of metric over time for the selected population

 **create_itsa()**
metric, percent
Runs interrupted time-series analysis to estimate the impact of an intervention at a specified time point.

 **create_rogers()**
metric, percent
Generates a Rogers adoption-curve style view of adoption using a usage metric/segment distribution.

 **xicor()**
metric, percent
Computes Chatterjee's Xi (rank correlation) to quantify association between two variables (including non-linear relationships).

Other analysis

 **create_lorenz()**
This function computes the Gini coefficient and plots the Lorenz curve based on a selected metric from a Person Query data frame

 **create_IV()**
Specify an outcome variable and return IV outputs

 **IV_report()**
The function generates an interactive HTML report using Standard Person Query data as an input. Report based on running the Information Value (IV) algorithm

 **maxmin()**
This function allows you to scale vectors or an entire data frame using the max-min scaling method, always returning a numeric vector

Network analysis

Analyze edge list datasets (e.g., Person-to-person, Group-to-group) from Viva Insights

 **network_g2g()**
Pass a data frame containing a group-to-group query and return a network plot

 **network_p2p()**
Pass a data frame containing a person-to-person query and return a network visualization

 **network_summary()**
Pass an igraph object to the function and obtain centrality statistics for each node in the object as a data frame

Helper functions

 **anonymise(), anonymize()**
Anonymize categorical variables such as HR variables by replacing values with dummy team names such as 'Team A'. The behavior is to make 1 to 1 replacements by default

 **totals_bind()**
The purpose of this is to enable to creation of summary tables with a calculated "Total" row

 **totals_col()**
Create a 'Total' column of character type comprising exactly of one unique value

 **tstamp()**
This function generates a time stamp of the format "ymmdd_hhmmss". This is a support function and is not intended for direct use

 **us_to_space()**
Convenience function to convert underscores to space

 **wrap()**
This function adds a character at the start and end of a character string, where the default behavior is to add a double quote

 **wrap_text()**
Wrap text in visualizations according to a preset character threshold

